	ESP gas lock	PE log curve	Sensor signals
Shape and schema	✓ 3,243 rows × 9 columns	✓ 2,922 rows × 7 columns	✓ 3,580 rows × 12 columns
Target and balance	✓ GAS_LOCK = 1 in 37.7% of rows	✓ PE 0.20–8.09, median 3.51	! labels −1/1 (57.6% = 1): map to 0/1
Missing values	! 300 cells, all in 50 esp_39 rows	✓ none	✓ none
Duplicates	! 2 identical rows; 6 repeat stamps	! 15 rows; a 12-row block repeats	✗ 2,484 rows repeat a feature vector
Keys and joins	✓ uuid unique in test; join on uuid	! no key: labels align by position	✓ uuid unique; labels join on uuid
Groups and order	✓ 7 entities, hourly TimeSlot	✗ no well ID; 7 depth resets	! no groups; uuid order has long runs
Units and meaning	! units undocumented	! units undocumented	! meaning and units undocumented
Test file inputs	✓ 3 unseen entities, same dates	✓ 310 rows, Depth 2964.5–3122.5	✗ 646 of 895 repeat a training row